Charan Bandi *Sr Software Engineer*

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Skills

Programming Languages & Frameworks
Python, Go, C/C++, Java, Spring Boot/Spring, Ruby on Rails, Node.js, React, JavaScript, TypeScript

Networking & Security
IPsec/IKEv2, VPN Tunneling (StrongSwan, Libreswan, OpenVPN, WireGuard), TLS/SSL, TCP/IP, DNS, HTTP/HTTPS, Zero Trust Architecture, Network Load Balancing, PKI/Certificate Management

API Design & Integration
API Design, REST APIs, GraphQL, Webhooks, JWT/OAuth2 Authentication, OpenAPI (Swagger)

AI & Machine Learning
AI Systems, LLM Integration, RAG Pipelines, AI-Driven Automation, NLP, Security-Focused AI, PII/PCI Detection, TensorFlow, Scikit-learn

Cloud & DevOps
AWS, GCP, Azure, Docker, Kubernetes, Jenkins, TeamCity, GitHub Actions, Terraform, Terragrunt, Ansible, Puppet, Linux Systems Administration, Shell Scripting

Databases & Data Technologies
PostgreSQL, Redis, DynamoDB, MySQL, MongoDB, Oracle SQL, Databricks

Monitoring & Infrastructure Tools
Sumo Logic, Datadog, Zabbix, Nagios, Logstash, Grafana, Prometheus, PagerDuty

Development Tools & Collaboration
Git, Bitbucket, Jira, Confluence, Agile, Microservice Architecture, Selenium, JUnit, Power BI

Professional Experience

Gen Digital — 05/2023 – Present | California, United States
Senior Software Engineer
- Led architectural design and mentoring within the VPN team, driving technical redesigns of IPsec/IKEv2 tunnel management, VPN gateway configuration (StrongSwan/Libreswan), and routing algorithm optimization for Freemium/Premium user segregation, improving performance, connection reliability, and cost efficiency.
- Integrated backend systems across Norton, Avast, Avira, and SurfEasy VPN platforms, managing 5,000+ servers across 100+ global locations serving millions of users, owning VPN tunnel orchestration, certificate lifecycle management, and secure key exchange across multi-cloud and on-prem infrastructure.
- Architected and deployed scalable backend services using Python/Go on AWS and GCP, building critical cloud security features including dynamic VPN server provisioning, configuration management, and automated failover systems.
- Led migration of all Linux VPN servers from AMD64 to ARM64 architecture, delivering $168K in annual savings. Orchestrated infrastructure provider migrations, reducing costs by $15K/year, improving VPN connection speeds by 7%, and lowering server error rates by 37%.
- Built AI-driven systems for production support and problem management, including automated incident triage (Goalie) and root-cause analysis tooling that reduced mean-time-to-resolution across the VPN platform.
- Internal AI hackathon finalist for architecting a Private VPN layer for LLM interactions, enabling automatic PII/PCI detection and filtering at the network edge to protect consumer data throughout AI-powered service workflows.
- Supported the Gen Payments team managing over $5 billion in yearly e-commerce revenue by designing and implementing Java/Spring Boot integrations with external Payment Service Providers. These integrations facilitated subscription licensing, customer onboarding, and reconciliation processes across all Gen Brands (Norton, Avast, AVG, LifeLock) product lines.

NortonLifeLock — 07/2021 – 05/2023 | Virginia, United States
Software Engineer
- Designed and built cloud-native distributed systems in Python and Go, powering security services deployed across AWS and Azure environments with high availability and multi-region resilience.
- Developed and maintained secure, scalable CI/CD pipelines and deployment automation using Jenkins, TeamCity, and GitHub Actions, improving system reliability and shortening release cycles.
- Built and maintained REST APIs, serverless components (AWS Lambda), and database models (PostgreSQL, Oracle SQL, Redis) serving multi-region traffic for Norton Secure VPN and SurfEasy VPN backend systems.
- Led end-to-end delivery of high-impact features for identity and security products — owning design, implementation, testing, and production rollout — while architecting backend systems with efficient load distribution and security logic that significantly reduced system downtime.
- Collaborated with DevOps, SRE, and technical support teams to troubleshoot complex production issues and improve observability, while mentoring junior engineers through code reviews to elevate coding standards and cloud best practices.

George Mason University — 01/2020 – 06/2021 | Virginia, United States
Research Assistant
- Developed and deployed ML-driven applications using Python (NLTK, spaCy, Scikit-learn, Keras) for NLP analysis, data mining, and web scraping on remote Linux servers, processing and analyzing over 200,000 datasets for IoT security research.
- Contributed to 3+ major research projects and 2 peer-reviewed IEEE publications on hardware vulnerability analysis, collaborating with Dr. Setareh Rafatirad and a team of postdoctoral and graduate researchers.

- Led project implementation and deployment across multiple teams, ensuring timely delivery aligned with research objectives and publication schedules.
- Designed scalable data processing pipelines that analyzed and visualized large datasets using Python and BeautifulSoup/Selenium, significantly enhancing the efficiency of research workflows.

George Mason University 08/2020 – 04/2021 | Virginia, United States
Graduate Teaching Assistant
- Facilitated the "IT Architecture Fundamentals" course by grading 100+ assignments and exams, hosting weekly office hours, and responding to student queries.
- Contributed to curriculum design for undergraduate Information Technology courses, aligning academic materials with industry-relevant learning outcomes.
- Delivered detailed, constructive feedback to improve student understanding and course performance across 80+ enrolled students.

Cyient Ltd 03/2018 – 08/2018 | Hyderabad, India
Software Engineer Intern
- Built a Resource Forecasting System using C# and Xamarin for cross-platform deployment, streamlining resource allocation across 10+ internal projects.
- Processed and analyzed organizational datasets using RESTful APIs, enabling data-driven project customization and resource planning.
- Developed internal dashboards and analytics tools, supporting real-time visualization and decision-making within operational teams.

Education

Master of Science in Computer Science Virginia, United States
George Mason University - Volgenau School of Engineering

Bachelor of Technology in Computer Science (B.Tech) Hyderabad, India
Gandhi Institute of Technology and Management

Certifications

AWS Certified DevOps Engineer - Professional
Date issued: April 25, 2025 Expires: April 25, 2028 Validation Number: f1e8e8779df69d18f597cb5f885d

Publications

Automated Supervised Topic Modeling Framework for Hardware Weaknesses 2023
2023 IEEE 24th International Symposium on Quality Electronic Design (ISQED)

Ontology-Driven Framework for Trend Analysis of Vulnerabilities and Impacts in IoT Hardware 2021
2021 IEEE 15th International Conference on Semantic Computing (ICSC)

Projects

NVD and CWE Vulnerability Security
Python, NLTK, spaCy, Scikit-learn, Keras, Selenium, BeautifulSoup, PyQt5
- Engineered a Natural Language Processing (NLP) system using NLTK and spaCy, achieving 95% accuracy in Part-of-Speech (POS) tagging, with a PyQt5 GUI for IoT security that processed and analyzed over 200,000 datasets. Employed advanced web scraping techniques using BeautifulSoup and Selenium to identify patterns within corpus descriptions and uncover meaningful associations.
- Developed Machine Learning (ML) models leveraging algorithms such as Time Series Prediction, Auto Regression, and LSTM RNNs to accurately forecast features including vector magnitudes, impact levels, and potential future security threats. This project was selected for publication at two IEEE conferences for its contribution to IoT hardware vulnerability analysis.

Housing Analytics
Java, Android, Node.js, MongoDB, REST APIs
- Developed a data-driven Java application targeting U.S. real estate challenges, comprising over 20,000 LOC and integrating 30+ third-party libraries. Improved housing data visualization and analytics for a dataset of 1.5 million properties on the Android platform.
- Designed and implemented robust MongoDB databases using advanced Mongoose schemas. Built complex query pipelines and integrated machine learning algorithms to enable accurate housing price predictions and automated feature extraction.